

Day 28 — Fri 2 Oct

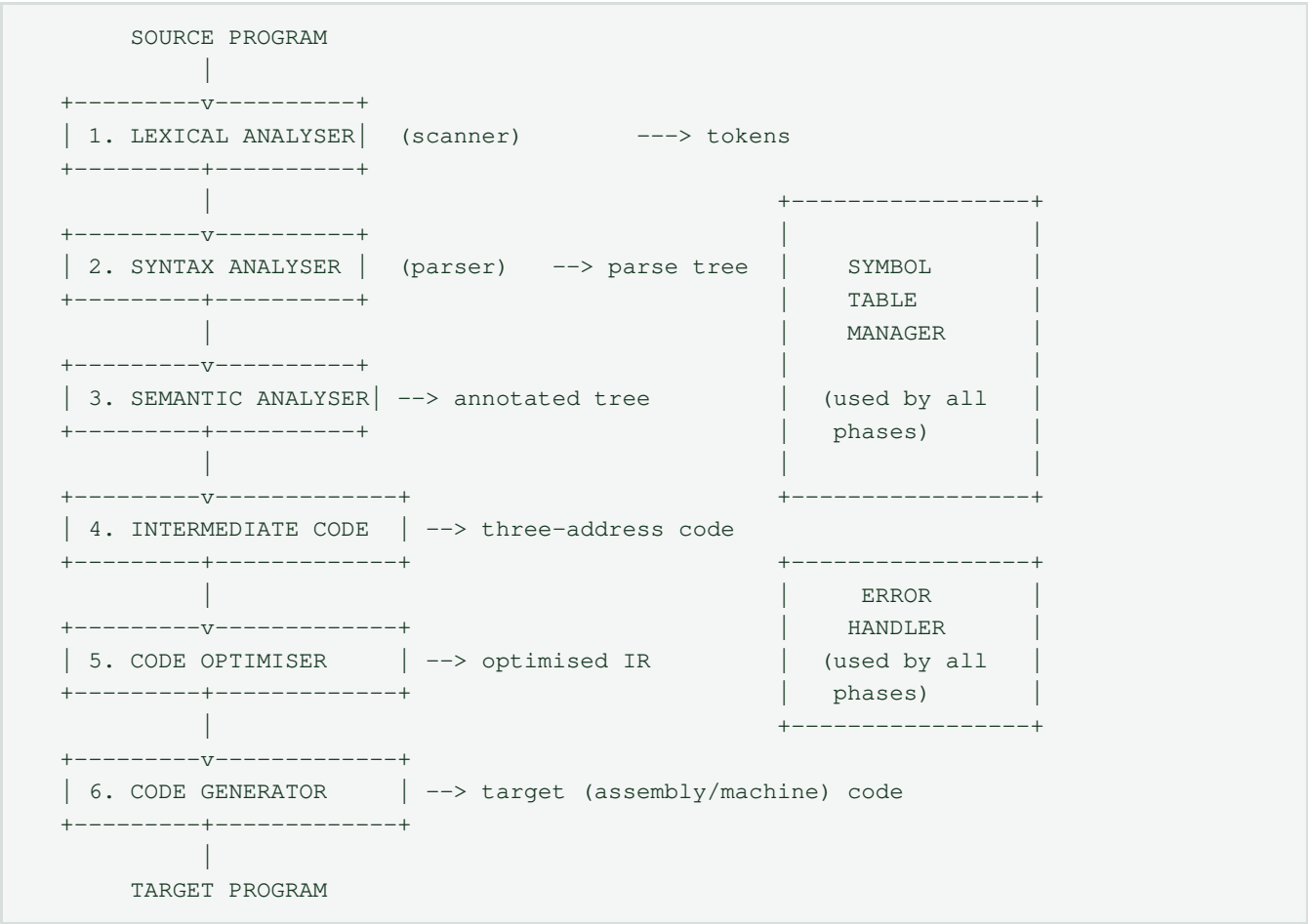
Countdown: 8 days to CS Degree Paper-I

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

● Degree — compiler phases

Source: ComputerScience_Degree/CSD_02_SystemSoftwareAndCompilers.md

The six phases — the diagram to reproduce



(Draw the six boxes vertically in a column; draw the symbol table and error handler as two boxes on the side connected to all phases. That side-connection is worth a mark on its own — say "both are used by every phase".)

One statement traced through every phase

Trace the single statement: `position = initial + rate * 60`

Phase	What it does	Output for this statement
1. Lexical analysis	Group characters into tokens (lexeme, token-class); enter identifiers in symbol table; discard whitespace	<code>id1 = id2 + id3 * 60</code> where id1=position, id2=initial, id3=rate, and =, +, *, 60 (number) are tokens

2. Syntax analysis	Check tokens against the grammar; build a parse/syntax tree	Tree: = at root, <code>id1</code> left, <code>+</code> right; under <code>+</code> : <code>id2</code> and <code>*</code> ; under <code>*</code> : <code>id3</code> and <code>60</code>
3. Semantic analysis	Type-check; insert coercions. <code>60</code> is int, <code>rate</code> is real → insert <code>inttoreal</code>	Tree with <code>inttoreal(60)</code> inserted before the <code>*</code>
4. Intermediate code	Emit three-address code	<code>t1 = inttoreal(60) → t2 = id3 * t1 → t3 = id2 + t2 → id1 = t3</code>
5. Code optimisation	Simplify. <code>inttoreal(60)</code> is compile-time constant → fold; drop the redundant temporary	<code>t1 = id3 * 60.0 → id1 = id2 + t1</code>
6. Code generation	Emit target assembly, allocate registers	<code>LDF R2, id3 / MULF R2, R2, #60.0 / LDF R1, id2 / ADDF R1, R1, R2 / STF id1, R1</code>

This trace is the heart of the 15-mark answer. Learn it cold: state each phase, its *task*, and its *output* on this one statement. Add the two side modules (symbol table, error handler) and a one-line front-end/back-end grouping, and it is a full-marks answer.

Forensic — physical evidence & chain of custody

Source: ComputerForensic/FOR_01_ForensicScienceFundamentals.md

4.2 Types of physical evidence

Two useful classifications — give **both** in a 5-mark answer.

(a) By nature/origin

Category	Examples
Biological	Blood, semen, saliva, urine, hair, tissue, bone, botanical material
Chemical	Drugs, poisons, explosives, petroleum products, paints, dyes, fibres
Physical	Glass, soil, metal, tool marks, tyre and footwear marks, firearms and bullets
Documentary	Handwriting, signatures, printed and typed documents, currency
Impression / pattern	Fingerprints, footprints, tool marks, tyre marks, bite marks, bloodstain patterns
Electronic / digital	Computers, hard disks, mobile phones, memory cards, CCTV footage, logs, cloud data
Miscellaneous	Anything else — clothing, jewellery, rope, cigarette ends

(b) By evidential value (the important one)

Type	Meaning
Transient evidence	Temporary; lost quickly — odour, temperature, wet stains, smoke. Record first.
Pattern evidence	Produced by physical contact — bloodstain patterns, glass fracture, tyre marks
Conditional evidence	Produced by an event/action — lights on/off, doors locked, TV channel, body position
Transfer / trace evidence	Material exchanged between persons/objects — hair, fibre, soil, paint
Associative evidence	Personal property linking a suspect to a scene — wallet, ID card, phone

Digital note for this elective: digital evidence is simultaneously *transient* (RAM, network connections), *transfer* (log entries created by an intrusion) and *associative* (a phone at

the scene). Say this in the exam — it shows you connected the two halves of the syllabus.

4.3 Class characteristics vs individual characteristics — the classic question

The intuition first

Ask: "How many objects in the world could have produced this evidence?"

- If the answer is "**a whole group / batch / model**" → the feature is a **class characteristic**.
- If the answer is "**only this one specific object, and no other in existence**" → it is an **individual characteristic**.

Class characteristics come from the **design/manufacture** of the object — they are shared by every item made to the same specification. Individual characteristics come from **random events** that happened only to that one object: manufacturing imperfections, wear, damage, growth, or random biological variation. Randomness is what creates uniqueness.

Formal definitions

- **Class characteristic:** a property of physical evidence that can be associated only with a **group** (a class) of objects or persons, never with a single source. It permits **identification** (this is glass; this is a Bata sole, size 9) and **exclusion**, but never individualisation.
- **Individual characteristic:** a property that can be attributed to a **single, unique source** with an extremely high degree of certainty. It permits **individualisation**.

Comparison table (reproduce this)

Aspect	Class characteristic	Individual characteristic
Source	Design / manufacture / species	Random wear, damage, imperfection, random biology
Points to	A group of possible sources	One source
Conclusion possible	"Consistent with / cannot be excluded"; exclusion	"Identified to the exclusion of all others"
Evidential weight	Corroborative	Conclusive / individualising
Statistical basis	Frequency of the class in the population	Practical uniqueness
Result of a mismatch	Exclusion (a mismatch in class characteristics eliminates the source outright)	Exclusion

Examples — learn six of each

Evidence	Class characteristic	Individual characteristic
Footwear	Brand, model, size, sole tread design	Cuts, nicks, wear pattern, embedded stone in the sole
Fingerprint	Pattern type — loop, whorl, arch; ridge count	Minutiae / Galton details — ridge endings, bifurcations, their relative positions
Firearm/bullet	Calibre, number of lands and grooves, direction and width of twist	Striations from tool marks unique to that barrel

Tool marks	Type and width of the tool (screwdriver, 12 mm)	Striae from nicks and imperfections on that tool's edge
Blood	ABO group, Rh type, species of origin	DNA STR profile (except identical twins)
Hair	Species, race, body area, colour, treatment (dye)	Nuclear DNA from the root sheath (the shaft alone gives only mitochondrial DNA → maternal lineage, still not individualising)
Glass	Type (float, tempered), colour, thickness, refractive index , density	Physical fit of a fracture edge — jigsaw match
Paint	Layer sequence, colour, chemical composition, make/model of vehicle	Physical fit of a paint chip to the damaged area
Handwriting	Copybook style, general slant	Individual writing habits, letter formations, pen lifts
Fibre	Type (cotton/nylon), colour, dye, cross-section	Almost never individualising (physical fit of a torn fabric edge is the exception)
Typewriting / printer	Typeface, font, model of machine	Defects — worn or misaligned characters, printer banding/ artefacts
Digital analogue	File type/extension , MIME type, camera model from EXIF	Hash value (SHA-256) , MAC address, IMEI, serial number, GUID

Key insight to write down in the answer

A **mismatch in class characteristics is conclusive of exclusion**, even though a *match* in class characteristics is not conclusive of identity. This asymmetry is why class evidence is still valuable: it eliminates suspects cheaply.

Also: enough class characteristics *in combination* can approach individualisation statistically (e.g. blood group A + Rh– + a rare enzyme variant narrows the population to a fraction of a percent) — but it is **cumulative probability**, not true individualisation.

Likely exam framing

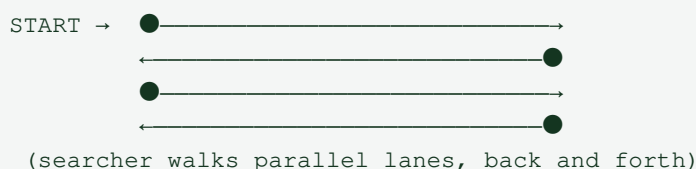
- 5 marks: "Distinguish between class and individual characteristics with examples." → table + 4 examples.
- 10 marks: add the exclusion asymmetry, the cumulative-probability point, and a digital analogue.

4.4 Search methods for locating physical evidence at the scene

Concept: a scene search must be **systematic, exhaustive and documented**. The method chosen depends on the **size** of the area, the **terrain**, the **number of searchers** available and whether the area is **indoor or outdoor**. Never search randomly; never let searchers cross uncleared ground.

Draw all five patterns. Each is a simple sketch — practise until you can draw all five in 90 seconds.

(1) Strip / Line (Lane) search



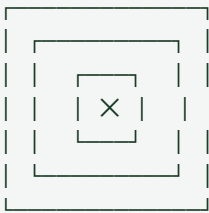
- The area is divided into **parallel strips/lanes**. One or more searchers walk each lane end to end, then turn and take the adjacent lane.
- With several searchers they stand in a **line, arm's length apart**, and advance together on a command ("line search").
- **Best for:** large **open outdoor** areas — fields, roadsides, parks; also for searching a route for a discarded weapon.
- **Advantages:** simple, easy to supervise, scales to many searchers.
- **Limitation:** in tall vegetation, items can be stepped over.

(2) Grid (Double-strip) search


 pass 1: lanes north-south
 pass 2: lanes east-west (over the same ground)

- **A strip search performed twice, the second time at 90° to the first.** Each point of ground is therefore examined from two different directions and lighting angles.
- **Best for:** large outdoor areas where thoroughness matters more than speed — bomb-blast sites, aircraft crash sites, searching for a small item like a cartridge case.
- **Advantage:** the **most thorough** of all patterns; a second look from a different angle finds what the first missed.
- **Limitation:** roughly **twice the time and manpower**.

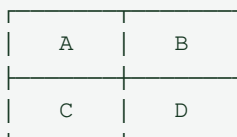
(3) Spiral search (inward or outward)



X = focal point (body / point of impact)

- The searcher moves in a **circular/spiral path**, either **outward** from a central focal point (the body, the seat of the fire) to the perimeter, or **inward** from the perimeter to the centre.
- **Inward spiral** is used when the centre is the most important area and you must avoid disturbing the periphery on the way in; **outward spiral** is used when the investigator is already at the centre (e.g. arriving at a body).
- **Best for: small, confined or barrier-free areas**, a single body in the open, **underwater** searches by a diver on a fixed line, and large open areas with no reference features.
- **Advantage:** good for a **single searcher**; simple around a focal point.
- **Limitation:** hard to maintain an even spiral; risk of missed gaps; not easily supervised.

(4) Zone / Quadrant / Sector search



Each zone assigned to a team;
zones may be sub-divided (A1, A2 ...)

- The scene is divided into **zones/quadrants**, each assigned to a searcher or team; each zone

may be sub-divided and searched using any of the other patterns internally.

- **Best for: indoor scenes** — a house is naturally divided into rooms; also large complex outdoor scenes and multi-storey buildings.
- **Advantage:** clear allocation of responsibility and accountability; allows **cross-checking** (teams swap zones for a second pass); scales well.
- **Limitation:** boundaries between zones can be neglected; requires coordination.

(5) Wheel / Ray / Pie search



Searchers move outward from the centre along radii ("spokes"), then return

- Searchers begin at a central point and move **outward along radii (spokes/rays)**, like the spokes of a wheel; or move inward along the spokes.
- **Best for: small, circular or roughly circular** scenes with an obvious focal point — e.g. a body in an open field, a small explosion crater.
- **Advantage:** fast; focuses effort near the centre.
- **Limitation: major gaps develop as the searchers move outward** (the spokes diverge) — so it is *not* thorough for large areas. This limitation is a favourite MCQ point.

Summary table (reproduce this)

Pattern	Best used for	Searchers	Thoroughness	Key weakness
Strip / Line	Large open outdoor areas	Many	Good	Single viewing angle
Grid	Critical outdoor scenes; small items	Many	Highest	Time and manpower ×2
Spiral	Small confined areas; single body; underwater	One	Moderate	Gaps; hard to keep even
Zone / Quadrant	Indoor scenes, buildings, complex scenes	Teams	Good, cross- checkable	Zone boundaries missed
Wheel / Ray	Small circular scenes with a focal point	Few	Poor at the periphery	Diverging gaps

General rules for any search

- Search **from the general to the specific**, and from the **periphery inward to the body** where approach paths must be preserved.
- Establish a **single common entry/exit path** and search it first.
- Photograph and document *before* anything is moved.
- Search **three-dimensionally** — floor, walls, ceiling, above eye level, under furniture, drains.
- **Search twice** — a second search by different personnel finds what the first missed.
- Note **negative evidence** too (the absence of something that should be there — no blood where there should be, a missing knife from a block, a wiped hard disk).

4.5 Chain of Custody — guaranteed exam material

Concept

The court will only accept an exhibit if the prosecution can prove that the object produced in court is **the very object collected at the scene**, and that it was **not altered, substituted, tampered with or contaminated** at any point in between. The chain of custody is the documentary proof of that.

Definition: The **chronological documentation** (paper and/or electronic) recording the **seizure, custody, control, transfer, analysis and disposition** of physical or electronic evidence — i.e. an unbroken record of *who* had the evidence, *when*, *where*, *why* and *what they did to it*, from the moment of collection until its production in court and final disposal.

Also called the "**paper trail**" or **chain of evidence**.

Why it matters

1. Establishes **authenticity** — the exhibit is what it is claimed to be (Evidence Act / BSA requirement of proving a document/object).
2. Establishes **integrity** — it has not been altered.
3. Establishes **accountability** — a named person is answerable for every period of custody.
4. Prevents **contamination, substitution and planting** allegations.
5. Enables **reconstruction of handling** if a problem is later discovered.
6. It is the **first thing the defence attacks** in cross-examination.

What the chain-of-custody form records

Field	Detail
Case / FIR number, police station, section of law	Identifies the case
Unique exhibit number	e.g. "Ex-A/3"
Description of the item	Make, model, colour, serial number/IMEI , capacity, visible damage
Date, time and exact place of collection	GPS/room/position
Name, designation and signature of the collector	Who seized it
Names and signatures of independent witnesses (<i>panchas</i>)	The <i>panchnama</i> /seizure memo requirement in India
Method of collection and packaging	Container type, sealing details, seal number/impression
Hash value (for digital evidence)	MD5/SHA-256 of the acquired image — the digital equivalent of a seal
Every transfer : released by (name, sign, date, time) → received by (name, sign, date, time), and purpose	The core of the chain
Storage location and conditions	Malkhana/property room, refrigerated, sealed cupboard, safe
Examination record	Who examined it, when, what was done, whether the seal was intact on receipt and re-sealed after
Final disposition	Returned, retained, destroyed (with court order)

The rule in one line

Every person who handles the evidence must sign for it, and there must be no unexplained gap

in time or custody between one signature and the next.

How the chain breaks — and the consequence

How it breaks	Consequence
An unexplained gap in the record (item unaccounted for for hours/days)	Defence argues opportunity for tampering
A broken, missing or mismatched seal on receipt at the laboratory	Report may be rejected outright
No independent witnesses to the seizure / defective <i>panchnama</i>	Seizure itself challenged
Missing signature at a transfer	That link cannot be proved
Evidence left unattended or in an unsecured place (a car boot, an open desk)	Integrity in doubt
Improper storage — biological sample not refrigerated, drive near a magnet	Degradation; result unreliable
Original examined instead of a working copy (digital)	Alteration of the original; hash mismatch
Hash mismatch between acquisition and analysis	Conclusive proof of alteration — evidence discarded
Absence of the §65B / BSA §63 certificate for electronic evidence	Inadmissible (see §6)
No record of who opened and re-sealed the packet	Contamination/substitution argument

Legal consequence: a broken chain does not automatically make evidence inadmissible in India in every case, but it **destroys its evidentiary weight**; courts routinely discard exhibits where the chain is doubtful, and in the case of electronic records the statutory certificate requirement makes the defect fatal. In the exam write: *"a break in the chain of custody renders the evidence liable to be excluded, or at minimum deprives it of evidentiary value, and can collapse the entire prosecution case."*

Digital-specific chain of custody

- Record the **hash at acquisition** and re-verify at every subsequent stage; the hash is the digital seal.
- Use **write blockers** so that the mere act of examination cannot alter the original.
- Work on a **verified working copy**, never the original.
- Maintain **tamper-evident, anti-static, Faraday** packaging (Faraday bag for live phones, to prevent remote wipe and network changes).
- Photograph** the device, its screen state, its ports and its serial/IMEI before seizure.
- Log every tool used, with version number.

§4.6 Likely exam questions (Unit 2)

Marks	Question
5	Define physical evidence and state its importance in criminal investigation.
5	Distinguish between class and individual characteristics with examples.
5	Describe any three crime-scene search methods with diagrams.
5	What is chain of custody? Why is it important?

10	Describe the various searching methods for locating physical evidence at a scene of crime, with diagrams, stating the suitability of each.
10	Explain chain of custody. What does the chain-of-custody form record, and what are the consequences of a break in the chain?
15	Discuss physical evidence — its types, characteristics and value — and explain how its integrity is maintained from the scene to the court.

MCQ traps (Unit 2)

- **Grid = double strip**, and is the **most thorough**; **wheel/ray** is the *least* suitable for large areas.
- **Zone** search is the standard **indoor** method.
- Fingerprint **pattern type (loop/whorl/arch) is a CLASS characteristic**; minutiae are individual.
- Blood **group is class**, **DNA is individual**.
- Refractive index of glass = **class**; a **physical (jigsaw) fit** = individual.
- Chain of custody documents **possession**, not analysis results.
- A mismatch in class characteristics → **exclusion** is valid.

§5. Scene of Crime (TP-I Unit 3)

5.4 Processing of physical evidence — the full lifecycle

Learn this as a **pipeline**; it is an easy 10- or 15-mark answer.

DISCOVER → RECOGNISE → DOCUMENT → EXAMINE (in situ) → COLLECT → PRESERVE → PACKAGE → SEAL → LABEL → FORWARD → (LAB) ANALYSE → REPORT → STORE → PRODUCE IN COURT → DISPOSE

(a) Discovering physical evidence

Systematic **search** (§4.4) plus enhancement techniques: oblique/grazing **light** to reveal dust prints and impressions, **forensic light sources / alternate light source (ALS)** in various wavelengths with goggles to make body fluids, fibres and bruises fluoresce, **UV and IR** photography, **luminol / BLUESTAR** for latent blood (including washed-away blood), **ninhydrin/cyanoacrylate/powders** for latent fingerprints, metal detectors for buried metal, cadaver dogs, ground-penetrating radar for buried remains.

(b) Recognising physical evidence

The single most important skill and the commonest failure. It requires the investigator to know what is **relevant to this offence**. Ask: *what would have had to happen for this crime to occur, and what would that have left behind?* Reconstruct hypothetically, then look for the predicted traces. Beware of: evidence that looks like rubbish (a cigarette end, a chewing-gum wad, a discarded receipt), **negative evidence**, and evidence that is **not at the scene but on the suspect or victim** (fingernail scrapings, clothing, shoes).

(c) Examination at the scene

Preliminary/**presumptive** tests only — Kastle-Meyer/phenolphthalein for blood, acid phosphatase for semen, colour tests for narcotics, portable spectrometers. These are **screening** tests: they can **exclude**, and they guide collection, but they are **not confirmatory** and their result must never be reported as a conclusion. Confirmatory tests belong in the laboratory.

Do only what cannot be postponed.

(d) Collection

- Document (notes, photo, sketch) **first**; then collect.
- Collect the item **in its entirety** wherever possible — take the whole cushion, not a cut-out; take the whole door lock, not just the marks.
- Collect from **most fragile/transient first** (the crime-scene equivalent of the order of volatility).
- Where the item cannot be moved, collect the stain/trace: **swab** (moisten a sterile swab with distilled water for a dried stain), **scrape**, **tape-lift** (for fibres and hair), **vacuum** (last resort — collects too much), **cast** (dental stone for footwear/tyre marks, silicone for tool marks), or **lift** (adhesive lifter for fingerprints).
- **Always collect a CONTROL/reference sample**: unstained substrate next to the stain, soil from a nearby unrelated area, a known blood sample from the victim and suspect, a specimen of the suspect's handwriting. *Without a control the Principle of Comparison cannot be satisfied.*
- Use **clean, new** tools for each item; change gloves between items.
- Assign a **unique exhibit number** at the moment of collection and start the chain-of-custody entry immediately.

(e) Safety measures for evidence collection

Hazard	Measure
Biohazard (blood, body fluids — HIV, hepatitis B/C, TB)	Universal precautions : treat all body fluids as infectious. Double nitrile gloves , gown/coverall (Tyvek), N95/FFP2 mask , eye protection/face shield, shoe covers, hair cover
Sharps — needles, broken glass, knives	Never place hands where you cannot see; use forceps; puncture-proof containers ; never recap needles
Chemical — drug labs, explosives, corrosives, unknown powders	Respirator, chemical-resistant gloves, ventilation; do not open unknown containers; call the bomb squad/chemical team
Fire/explosion, structural collapse, electricity	Clear the scene until specialists declare it safe
Radiological/CBRN	Specialist teams only
Cross-contamination between exhibits	Change gloves and tools between items; package separately; separate personnel for suspect and victim/scene wherever possible
Contamination BY the investigator	Elimination samples (DNA and fingerprints) from all scene personnel; masks to prevent breath DNA
Digital-specific	Anti-static wrist strap and anti-static bags (not ordinary plastic — static discharge can destroy a drive); keep media away from magnets, radios and heat; Faraday bag for live phones; battery/charger for a live phone so it does not power down mid-seizure
General	Never eat, drink, smoke or apply cosmetics at the scene; wash hands after; dispose of PPE as biohazard waste

(f) Preservation

Keep the evidence in the condition in which it was found, for as long as it may be needed.

Evidence	Preservation requirement
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Wet biological stains (blood, semen)	Air-dry completely at room temperature in the shade before packaging; never dry with heat or sunlight
Dried blood/DNA samples	Cool and dry; refrigerate for medium term, freeze (–20 °C) for long term; keep away from humidity
Liquid blood (reference sample)	In an EDTA vacutainer, refrigerated (4 °C), never frozen
Viscera for toxicology	Preserved in saturated saline (and a separate portion in rectified spirit for some analytes); a control of the preservative must also be sent Δ verify the standard preservative combination against a forensic medicine text
Arson/fire debris	Airtight metal cans or nylon bags — accelerant vapours escape from ordinary plastic and paper
Volatiles / alcohol samples	Airtight, cool, with preservative (sodium fluoride + potassium oxalate for blood alcohol) Δ verify
Firearms	Unloaded and made safe, documented; do not clean; package to avoid disturbing residues; swab hands for GSR within a few hours
Documents	Flat, in transparent envelopes/folders; never fold along new lines; never staple, pin, or write on
Soil/glass/paint	Dry, in paper packets or vials
Impressions	Cast, or lifted and protected
Digital media	Anti-static bags; away from magnetic fields, heat, moisture and vibration; never in ordinary polythene; keep phones charged and network-isolated

(g) Packaging — the rules and the classic exam point

The golden rule: BIOLOGICAL EVIDENCE GOES IN PAPER, NEVER IN PLASTIC.

Why? Plastic is airtight; residual moisture cannot escape; bacteria and fungi grow and **degrade the DNA**. Paper "breathes" and allows the sample to continue drying.

Evidence	Correct packaging
Dried bloodstained clothing, biological items	Paper bags, paper bindles, cardboard boxes , breathable — after air-drying
Fibres, hair, glass fragments, paint chips, small trace	Paper bindle (druggist's fold) placed inside an envelope or vial
Wet/damp items	Air-dry first ; if impossible, transport in paper and dry at the earliest, documenting the fact
Arson debris / accelerants	Clean, unused metal paint cans (airtight), or special nylon/Kapak bags — NOT ordinary plastic (vapour loss and contamination)
Liquids	Leak-proof, non-breakable bottles with screw caps, sealed and placed inside a secondary container
Sharp items (knives, syringes)	Rigid puncture-resistant container, blade immobilised, with hazard labelling
Firearms	Rigid box, secured to prevent movement, muzzle direction noted, unloaded
Documents	Flat between card, in a transparent envelope/folder; unfolded, unstapled
Impression casts	Rigid box with padding, after full curing

Hard disks, memory cards, USB drives	Anti-static (ESD) bags, then a rigid box with padding; tamper-evident evidence tape
Mobile phones	Faraday bag / signal-shielding bag; if switched on, keep it powered (with a charger inside the shielded bag) or note the decision to power down; document battery level
CDs/DVDs	Protective sleeves/jewel cases, avoid scratching the data surface
Latent-print items	Handle by edges, package so surfaces are not rubbed; box, do not bag loosely

General packaging rules

1. **One item, one package.** Never mix exhibits — cross-contamination.
2. Package so that the item **cannot move** inside the container.
3. **Do not over-handle** — package at the scene, at the point of collection.
4. Package so that the item can be examined **without breaking the seal in the wrong place** where possible.
5. Include the **control samples** in separate packages.
6. **Suspect's and victim's items must be packaged, transported and stored separately**, and preferably handled by different personnel.

(h) Sealing

- The package is closed so that it **cannot be opened without visibly destroying the seal** ("tamper-evident").
- Traditional Indian practice: the packet is stitched/wrapped in cloth and sealed with **lac/wax bearing a distinctive seal impression**; the **specimen impression of the seal** is separately forwarded to the laboratory so it can verify that the seal on receipt matches.
- Modern practice: **tamper-evident evidence tape** signed and dated **across the seam**, so that the signature is broken if the tape is lifted.
- The **seal number/description is recorded** on the seizure memo and chain-of-custody form.
- The seal is applied **in the presence of independent witnesses**, who also sign.
- The laboratory records on receipt whether the **seal was intact and matched the specimen**; if not, this is noted in the report and is usually fatal to the exhibit.
- **Digital equivalent of a seal = the hash value.**

(i) Labelling

Every package must bear, on the package itself (not only on a tag that can fall off):

1. **Case/FIR number**, police station, sections of law
2. **Unique exhibit number** ("Ex-4")
3. **Description** of the contents
4. **Date, time and place of collection**
5. **Name and signature of the collecting officer**, and of witnesses
6. **Name of the person from whom / place from which** it was seized
7. **Seal details**
8. **Hazard warnings** (biohazard, sharps, flammable) where applicable
9. Any special handling instruction ("keep refrigerated", "fragile", "anti-static")

(j) Forwarding

- A **forwarding letter/requisition** from the investigating officer or the court accompanies the exhibits, stating: the case details, a **list of exhibits with their seals**, a brief **history of the case** (essential — the laboratory needs context to choose the right analysis), and the **specific questions to be answered** ("is this human blood and does it match Ex-7?").

- The **specimen seal impression** is enclosed.
- Exhibits are sent by a **responsible officer by hand**, or by an approved secure channel; never by ordinary post for critical exhibits.
- **Perishable exhibits are sent first and fastest**; refrigerated transport where required.
- The lab issues an **acknowledgement/receipt** noting the condition and seals — this becomes the next link in the chain of custody.
- The **court's order** may be required for forwarding certain exhibits ⚠ verify the relevant BNSS provision (formerly CrPC §§ relating to sending exhibits for examination).

§5.5 Likely exam questions (Unit 3)

Marks	Question
5	What is a scene of crime? Distinguish between primary and secondary scenes.
5	Describe the duties of the first officer at the scene of crime.
5	Distinguish between a rough sketch and a finished sketch.
5	Why must biological evidence never be packaged in plastic?
5	Write a note on 3-D scanning of a crime scene.
10	Describe crime-scene documentation under the heads of note taking, photography, videography and sketching.
10	Explain the methods of measurement used in crime-scene sketching, with diagrams.
10	Describe the processing of physical evidence from collection to forwarding to the laboratory.
15	Describe in detail the procedure to be followed at a scene of crime from the arrival of the first officer to the forwarding of exhibits to the laboratory, with particular reference to a scene involving computers.
15	Discuss crime-scene documentation and the packaging, sealing, labelling and forwarding of physical evidence. Why is each step legally significant?

MCQ traps (Unit 3)

- Photography sequence is **overall** → **mid-range** → **close-up**, and close-ups are taken **without, then with**, a scale.
- The **rough sketch** is made at the scene and is **not to scale**; the **finished sketch is to scale**. The rough sketch must be preserved.
- **Triangulation** = distances from **two fixed points**; **rectangular coordinates** = perpendicular distances from **two walls at right angles**; **baseline** = along-and-offset from a line.
- **Cross-projection sketch** is for evidence on **walls and ceiling**.
- Arson debris goes in **airtight metal cans**, not paper and not ordinary plastic — this is the one exception to "biological in paper", and examiners love it.
- Wet bloodstains are **air-dried at room temperature in the shade**, not heated, not sun-dried.
- A **presumptive test is not confirmatory**.
- The first responder's first priority is **safety/preservation of life**, not evidence.

§6. Indian Legal Framework for Electronic Evidence

(exam-appropriate depth; accuracy flagged where uncertain)

⚠ Read this whole section with care. Section numbers are the easiest place to lose marks by

confidently writing something wrong. Where a number is marked $\triangle$, verify it against indiacode.nic.in before the exam. **In the exam, when unsure of a number, describe the provision correctly and name the statute — a correct description without a number scores far better than a wrong number.**